

Scenario

Smart Lab Equipment & Slot Reservation Portal

Name : Navya R

SRN : PES2UG24CS308

Section : F

Lab : 3

Primary Domain: Campus & Academic Operations

Target Actors: Student, Lab Technician

STEP 1

Functional Requirements

1. Student can log in/register.
2. Student can view available lab equipment.
3. Student can check equipment availability.
4. Student can view available lab slots.
5. Student can reserve an available slot.
6. Student can cancel a reservation.
7. Lab Technician can add/update/remove equipment details.
8. Lab Technician can create/update/cancel lab slots.
9. Lab Technician can view reservations.

10. System can send reservation notifications.

Non-Functional Requirements

1. **Security** – Only authorized students and lab technicians should access the system.
2. **Performance** – Equipment and slot availability should load quickly.
3. **Usability** – The portal should be simple and easy to use.
4. **Reliability** – The system should prevent double booking.
5. **Availability** – The portal should be accessible when students need to reserve slots.
6. **Data Integrity** – Reservation and equipment information should remain accurate.

Main Challenges

Performance:

The system should quickly display available equipment and slots, especially when many students access it simultaneously.

Usability:

Students should be able to easily find equipment, check slots, and make reservations.

Security:

Only authorized users should be allowed to make reservations or manage equipment and slots.

The lab specifically asks you to identify functional/non-functional requirements and challenges involving performance, usability, and security.

STEP 2

Architectural Style Analysis

Architecture	Advantages	Disadvantages
Layered	Clear separation of concerns, easy to understand and maintain	Some performance overhead, layers can become dependent
Microservices	Independent scaling, fault isolation, independent deployment	More complex, network latency, data consistency issues
Client-Server	Centralized control, simple deployment, easy data consistency	Server can become bottleneck/single point of failure

Step 3

Select Architecture & Identify Components

Selected Architecture:

Layered Architecture

We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal.

Reason 1

It provides clear separation between the presentation, business logic, and data management parts of the system, making the portal easier to understand and maintain.

Reason 2

The architecture is suitable for handling different functions such as user management, slot reservation, equipment management, and database operations separately.

Security Advantage

Authentication and authorization can be handled in the appropriate application layer, helping ensure that only authorized students and lab technicians can access protected functions.

Performance Benefit

Separating responsibilities into layers allows requests such as slot availability and equipment information to be processed efficiently without mixing unrelated system operations.

5 Components are:

1. User Management Component

Handles:

- Student login
- Lab Technician login
- User information
- Authentication

2. Slot Reservation Component

Handles:

- Viewing slots
- Booking slots
- Cancelling reservations
- Reservation information

3. Equipment Management Component

Handles:

- Equipment details
- Equipment availability
- Adding/updating/removing equipment

4. Notification Component

Handles:

- Booking confirmation
- Cancellation notification
- Slot/equipment updates

5. Database Component

Stores:

- Student/Lab Technician data
- Equipment data
- Slot data
- Reservation data

Step 4:

Component Diagram Creation

Identified Components

1. **User Management Component** – Handles student and lab technician login, authentication, and user information.
2. **Slot Reservation Component** – Handles viewing available slots, booking slots, cancelling reservations, and reservation information.
3. **Equipment Management Component** – Manages equipment details, equipment availability, and adding, updating, or removing equipment.
4. **Notification Component** – Handles booking confirmations, cancellation notifications, and slot/equipment updates.
5. **Database Component** – Stores student/lab technician, equipment, slot, and reservation data.

Interfaces and Interactions

The components communicate through interfaces to exchange required information and services.

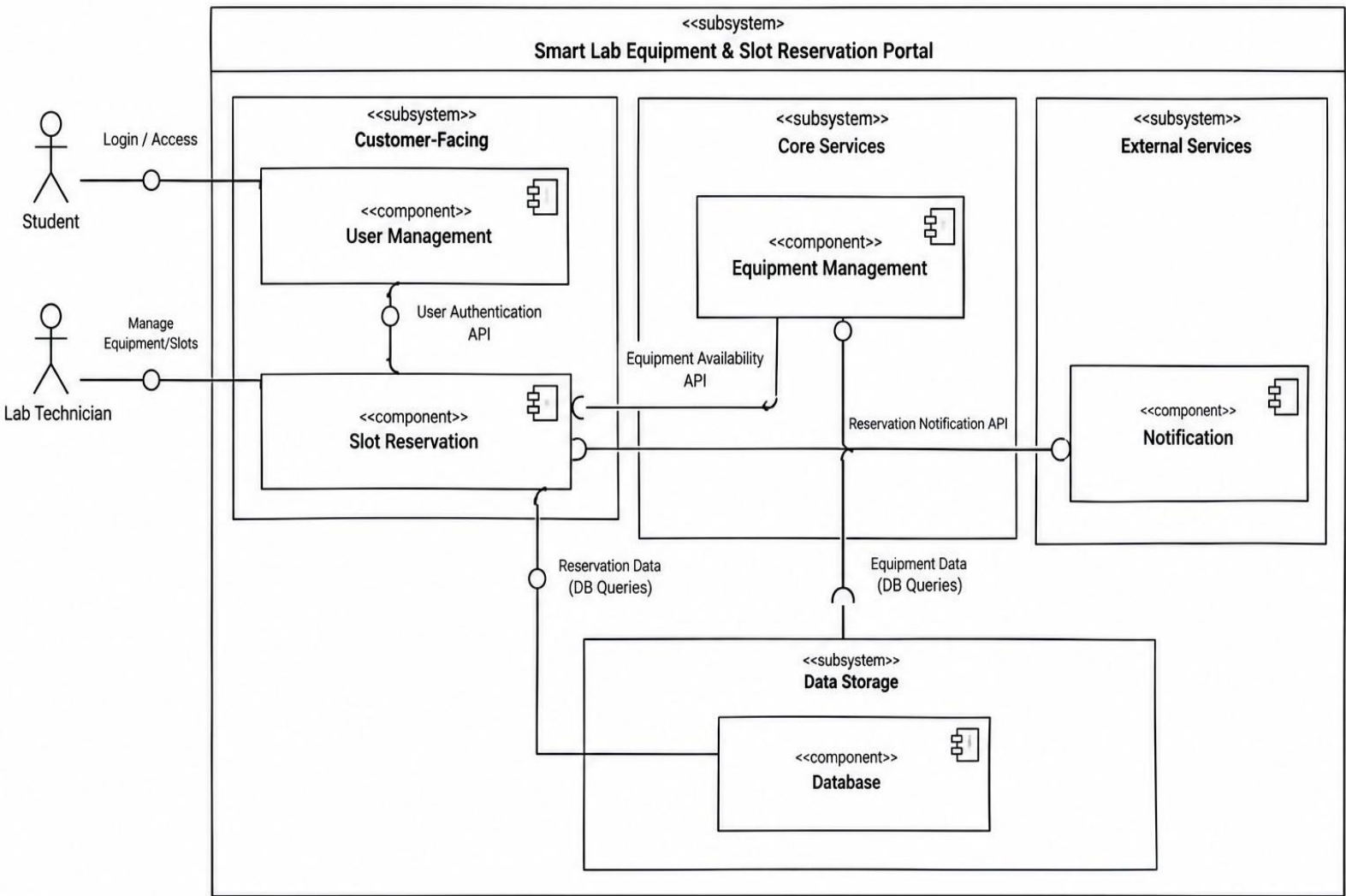
- **User Authentication API** – User Management provides authentication services to the Slot Reservation component.
- **Equipment Availability API** – Equipment Management provides equipment availability information to the Slot Reservation component.

- **Reservation Data (DB Queries)** – Database provides reservation data to the Slot Reservation component.
- **Reservation Notification API** – Notification provides notification services to the Slot Reservation component.
- **Equipment Data (DB Queries)** – Database provides equipment data to the Equipment Management component.

UML Notation Used

- Each system module is represented as a UML component using the <<component>> notation.
- Provided interfaces are represented using the ball (lollipop) notation.
- Required interfaces are represented using the socket notation.
- Solid lines represent communication between components.
- Interface names are labelled to show the service or data exchanged.

Component Diagram



Architecture Selection – Smart Lab Equipment & Slot Reservation Portal

Justification

Architecture Selection

We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal.

Architectural Choice

The selected architectural style is Layered Architecture, consisting of the Presentation Layer, Business Logic Layer, and Data Layer.

Two Reasons

- 1. Clear separation of responsibilities:** The portal can separately handle user management, slot reservation, equipment management, and database operations, making the system easier to understand and maintain.
- 2. Suitable for the portal's functions:** Student and lab technician requests can be processed through the appropriate layers, while reservation, equipment, and user data are handled by the data layer.

Security Advantage

Authentication and authorization can be handled in the appropriate application layer, helping ensure that only authorized students and lab technicians can access protected functions and data.

Performance Benefit

Separating responsibilities into layers allows requests such as slot availability and equipment information to be processed efficiently without mixing unrelated system operations.

Conclusion

Therefore, Layered Architecture is a suitable choice for the Smart Lab Equipment & Slot Reservation Portal because it provides clear organization, easier maintenance, security control, and efficient processing of system requests.